

DigitalTwin-Predict: Digital Twin-Based Kiln Health Monitoring

This plan outlines the implementation strategy for Idea 2, tailoring the workload perfectly to **Mubarak** (Software/Web/AI/Hackathon pro) and **Wale** (MechEng/Hardware Sim/AI scholar).

Team Work Breakdown (Mubarak & Wale)

Given your respective strengths, here is the optimal split for maximum impact and efficiency in the 7-day timeframe:

Wale (Mechanical Engineering, Hardware Simulations, AI/ML)

Wale will handle the core engineering realism, physics modeling, and simulation logic.

- Kiln Thermodynamics & FEM Concept:** Define the physical parameters, heat transfer mechanics, and failure modes (refractory wear) of a rotary cement kiln.
- Simulation Logic & Data Modeling:** Design the mathematical models that dictate how thermal stress leads to shell deformation over time. Wale will outline the rules for the synthetic data generator.
- Physics-Informed ML Model:** Work on the logic/math for the ML regression model that predicts Remaining Useful Life (RUL) based on thermal expansion and wear.
- Engineering Documentation:** Lead the writing of the Risk & Safety Assessment, Technical Assumptions, and the mechanical feasibility sections of the final document.

Mubarak (Software, Web Dev, Mobile Gaming, Data Analyst, AI/ML)

Mubarak will handle the software engineering, 3D graphics, UI/UX, and data pipelines to bring the twin to life.

- 3D Digital Twin (Three.js/WebGL):** Leverage mobile gaming/web dev skills to build an interactive, rotating 3D model of the kiln using React Three Fiber.
- Dashboard UI/UX & Web Dev:** Build the main web application (React + Vite). Implement a premium, hackathon-winning interface with real-time charts, heatmaps, and alert systems.
- Data Pipeline & ML Integration:** Write the `simulator.js` script (translating Wale's math into code) and integrate the ML anomaly detection to feed live mock data to the dashboard.
- Pitch Deck & Presentation:** Use hackathon experience to design a killer 10-slide executive pitch deck and format the final business case for the judges.

Proposed Technical Implementation

Frontend Dashboard Application (Vite + React)

- index.css & Design System:** Implement a premium, dark-mode industrial aesthetic. We'll use glassmorphism, vibrant alert colors (red/orange for hot spots), and smooth micro-animations.
- components/KilnDigitalTwin.jsx:** A React Three Fiber component rendering a rotating 3D cylinder. We will map a dynamic texture/heatmap onto the cylinder to represent the shell temperature distribution based on Wale's FEM concepts.
- components/MetricsPanel.jsx:** Line charts displaying simulated thermal stress over time and Remaining Useful Life (RUL) predictions.
- components/AlertSystem.jsx:** A sidebar showing recent anomalies, predicted failure dates (2-4 weeks out), and escalation workflows.

Data Simulation & ML Mock Backend (JavaScript)

To keep the prototype rapid and robust for presentation, we will mock the backend logic directly in JS.

- lib/simulator.js:** A utility to generate realistic time-series data for kiln thermodynamics, simulating gradual refractory wear and the sudden appearance of hot spots (coded by Mubarak, based on Wale's physics model).
- lib/ml-rul.js:** A mock ML regression function predicting Remaining Useful Life based on the simulated temperature distributions and historical degradation rates.

User Review Required

- Tech Stack:** We will use **React + Vite** and **React Three Fiber (Three.js)**. Is this stack good to go for Mubarak?
- Are you ready to proceed with this plan?** Once you approve, I will initialize the React project and we can start building the dashboard!

Verification Plan

- Run the local Vite dev server and review the 3D kiln visualization, ensuring the heatmap dynamically updates based on the simulated data.
- Verify the UI aesthetics feel premium and responsive.
- Validate that the dashboard accurately reflects the core concept of predicting failures 2-4 weeks in advance.